

PUBLIC DISCLOSURE EDITION

ODRE PQC v0.2.9

Public Technical Whitepaper Rev.2

English edition

2 September 2026

Contents

1. Executive summary
2. Product purpose and architecture
3. Cryptographic profile and security contract
4. Threat model and trust boundaries
5. Verification and audit evidence
6. Cross-platform results
7. Commercial layer separation
8. Deployment and disclosure boundary
9. Conclusion

Document identity

Product	ODRE PQC
Version	0.2.9
Artifact	odre_pqc-0.2.9-py3-none-any.whl
Wheel SHA-256	A8AFA10EE0AFACEF1C0FAF55FF07B96C722920E5AEE8692F6F026E804F028697
Evidence manifest SHA-256	4AF2C88FDEA0CB1FB030E4AB6D5B41394CC52397664364233482E3EC14945F3B
Cryptographic runtime	OpenSSL 3.5.8 / ML-KEM-768 / ML-DSA-65
Verified platforms	Windows Server 2022 AMD64; Ubuntu 24.04.4 LTS AMD64
Core audit verdict	ODRE_PQC_v0.2.9_FULL_DEVSEC_AUDIT_PASS

Executive summary

ODRE PQC is an application-embedded post-quantum security boundary for protected FastAPI routes. It combines cryptographic runtime management, authenticated session establishment, strict sequencing, replay resistance, protected request and response semantics, native artifact trust, lifecycle control and fail-closed enforcement behind a small integration surface.

Version 0.2.9 is the sealed cross-platform artifact identified by the SHA-256 in this document. The exact same wheel passed a fresh full development-security audit on Windows Server 2022 AMD64 and Ubuntu 24.04.4 LTS AMD64. No confirmed Critical, High or Medium product finding remained within the executed scope.

This paper reports evidence, not absolute security. PASS applies only to the named artifact, platforms, configurations, threat models and executed tests. Production deployment remains a separate operator-controlled decision.

Product purpose and architecture

ODRE PQC protects selected application routes. It does not replace TLS, identity, authorization, secure coding, host hardening, patching, backups or monitoring. Protected business handlers run only after the PQC boundary accepts the request.

The public integration model is intentionally small: initialize, diagnose, verify, call `install(app)`, start the server and inspect authenticated status. The wire contract is versioned as `odre-pqc/3`. Platform differences remain inside adapters and platform-specific native runtime bundles distributed through one Python wheel.

```
Integration flow: odre-pqc init -> doctor -> verify -> from odre_pqc import install; install(app) -> server  
start -> odre-pqc status
```

Cryptographic profile and security contract

The audited runtime identifies as OpenSSL 3.5.8 (OpenSSL_version_num 0x30500080). Real ML-KEM-768 and ML-DSA-65 self-tests and the hybrid handshake passed on both target operating systems.

ODRE PQC does not silently fall back to classical-only protection for protected routes. If provider, key material, session state, replay state, native trust or runtime health cannot satisfy the contract, the protected route is blocked before business logic runs.

Authenticated inner request semantics are the semantics observed by the handler. Protected response status, ordered headers, cookies and body remain inside the protected response frame. Replay, rollback, gap, mismatch, cross-route and cross-session ambiguity are rejected fail-closed.

Threat model and trust boundaries

The audit exercised network tampering, replay and concurrency, malformed and oversized inputs, semantic confusion, response metadata leakage, crash/restart state, local filesystem replacement, symlink or reparse manipulation, dependency/provider injection and lifecycle races.

The product assumes that the operating system, CPU, Python runtime and authorized administrator are not fully compromised. It does not claim protection against a malicious kernel, unrestricted administrator control, physical extraction, compromised admitted business logic, untested denial of service or future cryptanalytic breaks.

Verification and audit evidence

The sealed wheel was installed from a fresh copy under hash lock. Start and end hashes matched. Earlier reports were not reused as PASS evidence, and product bytes were not changed during the audit.

Static review covered 38 Python files and 7,160 lines plus package metadata, RECORD, platform support and both native manifests and archives. Wheel RECORD verification passed 45/45. Fresh advisory queries covered 19 dependencies with zero query errors and zero vulnerability records returned at audit time.

Metric	Windows	Ubuntu
Platform verdict	PASS	PASS
Replay-race requests	1,280	2,880
Double handler execution	0	0
Native load attempts	3,000	-
Atomic replacements observed	1,062	-
Junction create/remove cycles	3,000	-
Late-promotion parent swaps	-	600 / 600
Untrusted load / object misbinding	0	0
Unresolved Critical / High / Medium	0 / 0 / 0	0 / 0 / 0

Cross-platform results

Windows testing covered offline installation, initialization, second-init protection, diagnostics, verification, authenticated status, session transitions, replay race, semantic binding, ACL trust, shutdown lifetime, tamper attacks, crash recovery and native load stress.

Ubuntu testing covered initialization, negative diagnostics, real PQC, replay race, admission fuzzing, malformed and oversized inputs, HTTP smuggling shapes, response secrecy, WebSocket fail-closed behavior, native object trust, concurrent initialization, worker collision and graceful/forced restart.

Commercial layer separation

Licensing, payment, download, Trial and monitoring services are a Commercial control plane and are not part of the sealed v0.2.9 cryptographic Core. Server-side Commercial code passed 102/102 isolated tests, while 60/60 selected JigeumON regression tests passed. The isolated migration upgrade-downgrade-upgrade cycle passed.

These results are not a production deployment claim. PostgreSQL acceptance, live PayPal, SMTP, production secrets, trusted-proxy policy, deployed customer helper and end-to-end client entitlement enforcement were not completed in the cited Commercial monitoring report. Its verdict is code-and-isolated-tests PASS, production NOT READY.

Deployment and disclosure boundary

Officially verified environments are Windows Server 2022 AMD64 with CPython 3.12.10 and Ubuntu 24.04.4 LTS AMD64 with CPython 3.12.3. Other Linux distributions, Ubuntu 22.04, Debian 12 and ARM64 are not represented as verified by this evidence set.

No private keys, credentials, product source, wheel bytes, native binaries, test harnesses or customer data are disclosed in this public edition. Internal reports remain private. Numbers without a located standalone source report, including the previously referenced approximately 179.4-million fuzz lineage, are intentionally not claimed here.

Conclusion

ODRE PQC v0.2.9 provides a compact application integration surface backed by a strict fail-closed contract, real post-quantum algorithms and a sealed cross-platform artifact identity. The available evidence supports the stated development-security audit result within its defined scope.

Security remains a process. Operators must preserve artifact identity, follow supported deployment guidance, maintain the host and application, and treat future changes as new evidence obligations.

Public technical whitepaper Rev.2 - Core and Commercial claims are explicitly separated.